

Why Retrieving the Right Information Does Not Preserve the Right Context

Context Engineering, Retrieval-Augmented Generation (RAG), Semantic Retrieval, and Context Drift

Semantic Fidelity Project — Technical Mechanism Brief #3

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System Function

Modern AI systems increasingly depend on external knowledge.

Rather than relying only on what is stored in model weights, they retrieve documents, search vector databases, access memory stores, query knowledge graphs, and maintain state across long-running interactions.

This architecture is often described through terms such as Retrieval-Augmented Generation (RAG), context engineering, semantic retrieval, knowledge graph memory, and agent persistence.

These systems solve an important problem.

They allow models to work with information that exists outside the immediate context window.

Knowledge becomes easier to access.

Conversations become more continuous.

Agents can operate across longer time horizons.

As these systems improve, they appear increasingly informed.

Yet a deeper challenge emerges:

A system can retrieve exactly the right information while gradually losing the context that gives that information its meaning.

The challenge is not simply retrieving knowledge.

It is preserving context across retrieval.

Representational Chain

AI systems do not operate directly on knowledge.

They operate on representations of knowledge.

A document is a representation.

An embedding is a representation.

A vector is a representation.

A memory entry is a representation.

A knowledge graph relationship is a representation.

A context window is a representation.

Agent state is a representation.

These structures make large-scale retrieval possible.

Without them, long-horizon reasoning could not scale.

But every retrieval pipeline transforms information before the model ever reasons about it.

The chain is:

source knowledge → indexing → embedding → retrieval → context assembly → model
interpretation → output

At first, this relationship remains strong.

Retrieved information reflects the source.

The assembled context reflects the task.

The model remains grounded.

Over time, however, additional layers accumulate.

Knowledge is summarized.

Embedded.

Ranked.

Filtered.

Retrieved.

Reassembled.

Interpreted.

The retrieved representation survives.

The context that originally gave it meaning becomes progressively harder to preserve.

Failure Mechanism

Context drift usually emerges through four stages.

Storage

Information is collected and stored.

Documents, entities, memories, and state remain closely connected to the realities they represent.

Representation

Information is embedded, indexed, categorized, summarized, and structured.

These representations make retrieval efficient.

They also compress context.

Retrieval

The retrieval system successfully identifies information that appears relevant to the current task.

The original knowledge is now mediated through multiple representational layers.

Drift

The retrieval succeeds.

The memory loads.

The context window is populated.

Yet the relationship between the retrieved information and its original meaning gradually weakens.

The dominant mechanism is:

source knowledge → representation → semantic retrieval → context assembly → interpretation → context drift

The retrieved document may be correct.

The retrieved passage may be relevant.

The embedding similarity may be high.

Yet the model may misunderstand why the information matters, which assumptions surround it, what constraints originally applied, or how it should influence the present task.

The information survives.

The context changes.

Why the System Still Looks Successful

Context drift is difficult to detect because retrieval systems expose highly visible indicators of success.

The search returns relevant documents.

The vector database responds.

The memory store loads.

The context window fills.

The model references retrieved information.

Observers conclude that grounding has been preserved.

But retrieval and understanding are not identical.

A model can retrieve the correct document while emphasizing the wrong paragraph.

It can preserve the relevant facts while overlooking the assumptions that gave those facts significance.

It can assemble a complete context window while misunderstanding the relationships between its parts.

The visible infrastructure functions correctly.

The hidden failure occurs during interpretation.

The system appears grounded because the retrieval pipeline succeeded.

The model's understanding may still drift away from the context that retrieval was intended to preserve.

Technical Distinction

Retrieval Success

The requested information is successfully located.

Retrieval Relevance

The returned information appears related to the current query.

Context Persistence

Information remains available across sessions and workflows.

Context Fidelity

The retrieved information preserves the relationships, assumptions, priorities, constraints, and situational meaning necessary for correct reasoning.

These properties are not equivalent.

A retrieved passage may be relevant without being sufficient.

A complete context window may still omit what matters most.

A semantically similar document may not preserve the intended interpretation.

A grounded system may still generate an answer that departs from the context that grounded it.

This is why retrieval quality alone cannot guarantee reliable reasoning.

The source makes this distinction explicit.

Retrieval-Augmented Generation can successfully retrieve relevant documents, but the remaining challenge is ensuring that generated outputs stay faithful to those retrieved sources. Likewise, a context window may contain all necessary information, yet the model still interprets and prioritizes that information through its own representational processes.

Reality Drift Interpretation

Within the Reality Drift framework, retrieval systems are representational infrastructures.

They do not reconnect models directly to reality.

They reconnect models to representations of reality.

Those representations become the environment in which reasoning occurs.

As AI systems become increasingly dependent on retrieval, memory persistence, and external knowledge, preserving Semantic Fidelity becomes more important than retrieving larger amounts of information.

The challenge shifts.

It is no longer primarily a retrieval problem.

It becomes a representational problem.

The documents remain.

The embeddings remain.

The vector database remains.

The context window remains.

Yet the relationship between those representations and the realities they were intended to preserve may gradually weaken.

This is context drift.

The danger is not that the system lacks information.

It is that the system reconstructs the wrong situation from the information it successfully retrieved.

Recognition Questions

Why can RAG retrieve the correct document but still generate the wrong answer?

Because retrieval preserves access to information, not necessarily the context required to interpret that information correctly.

What is context drift?

Context drift occurs when retrieved representations remain available while the relationships that originally gave those representations meaning gradually weaken.

Why does semantic retrieval not guarantee understanding?

Semantic similarity measures representational closeness. It does not ensure preservation of assumptions, priorities, constraints, or intended interpretation.

What is the difference between retrieval relevance and context fidelity?

Retrieval relevance asks whether useful information was found. Context fidelity asks whether the retrieved information still supports the same meaning it had within its original context.

Why do larger context windows not eliminate context drift?

Larger windows increase available information but do not ensure that the model reconstructs the relationships that make that information meaningful.

Core Principle

Retrieval systems do not preserve knowledge directly. They preserve representations of knowledge.

A model can retrieve the correct information while reconstructing the wrong context from it.

The challenge is not accessing more information. It is maintaining fidelity between retrieved representations, their original context, and the reasoning they are meant to support.